PART 1

1.C
2.B
3.C
4.A
5.C
6.A
7.C
8.C
9.C
10.D
11.B
12.A

PART 2

13.

A. 1.2/0.4 = **3.0**
B. 400/3 = **133 N**
C. Friction of the fulcrum or weight of the lever arm

14.

a. 6x0.8 = **4.8**
b. 600/4.8 = **125 N**
c. The efficiency is less than 100 due to friction of pulleys and axles and weights of the rope

15.

a. 5/1.25 = **4.0**
b. 490/4 = **122.5 N**
c. **612.5 J** in both cases

16.

a. 120 x 4 = **480 J**
b. 360 x 1 = **360 J**
c. 360/480 = **75%**

17.

a. Possible answers are: **pulley + lever, inclined plane + pulley, screw + wheel and axle**
b. Combining simple machines multiplies mechanical advantage, reducing force but increasing distance
c. Increasing mechanical advantage increases the distance needed to apply force over to achieve the same amount of work, also can be increased friction, or increased complexity

18.

a. Tan 30
b. 3 x 0.5 = **1.5 N**

c. The angle would **not change**